

CSS Syntax and Selectors

Learn the structure of CSS rules and how selectors choose HTML elements to style.

■ CSS Syntax

To apply styles to HTML elements, you need to understand the basic syntax of CSS and how to select elements on the page.

Every CSS rule consists of a **selector** and a **declaration block**.

```
selector {  
  property: value;  
}  
  
h1 {  
  color: navy;  
  font-size: 32px;  
}
```

Part	Meaning
Selector	Targets the HTML element to be styled.
Property	Defines what aspect is being changed, such as color or font-size.
Value	Specifies the setting for a property, such as navy or 32px.
Declaration block	The declarations enclosed inside curly braces { }.
Semicolon ;	Each declaration ends with a semicolon.

■ Types of Selectors

Selectors determine which HTML elements a CSS rule applies to.

1. Element Selector

Selects all elements of a specific type. For example, this targets all **<p>** elements.

```
p {  
  color: gray;  
}
```

2. Class Selector

Selects elements with a specific class. Use a period . before the class name.

```
<p class="highlight">This is important.</p>  
  
.highlight {  
  background-color: yellow;  
}
```

3. ID Selector

Selects a single element with a unique ID. Use a hash # before the ID name.

```
<h1 id="main-heading">Welcome</h1>

#main-heading {
  font-family: Arial, sans-serif;
}
```

4. Universal Selector

Applies styles to all elements on the page. This is commonly used for resetting default styles.

```
* {
  margin: 0;
  padding: 0;
}
```

5. Grouping Selectors

Apply the same styles to multiple selectors at once. This avoids repetition.

```
h1, h2, h3 {
  color: darkblue;
}
```

6. Descendant Selector

Targets elements nested inside other elements.

```
<div>
  <p>This is a paragraph inside a div.</p>
</div>

div p {
  font-style: italic;
}
```

Only **<p>** tags inside **<div>** will be affected.

7. Combining Class and Element Selectors

You can be more specific by combining a class and an element selector. This targets only **<p>** elements with the class **note**.

```
p.note {  
  color: teal;  
}
```

■ Selector Quick Reference

Selector Type	Syntax	What It Targets
Element	p	All &lt;p&gt; elements.
Class	.highlight	Elements with the class highlight .
ID	#main-heading	The element with the specified ID.
Universal	*	All elements on the page.
Grouping	h1, h2, h3	Multiple selectors sharing the same styles.
Descendant	div p	&lt;p&gt; elements nested inside &lt;div&gt; .
Combined	p.note	Only &lt;p&gt; elements with class note .

■ Tip

- Use **.** for class selectors, **#** for ID selectors, and tag names for element selectors.
- Combine and group selectors when you need more precise or reusable styling.

Quick Takeaway

CSS selectors help you choose which HTML elements to style. Start with element, class, and ID selectors, then use universal, grouping, descendant, and combined selectors for more control.